Database System Engineering and Distributed Backend Development

WEEK-4: BookFlow Intelligent Catalog & API Blueprint

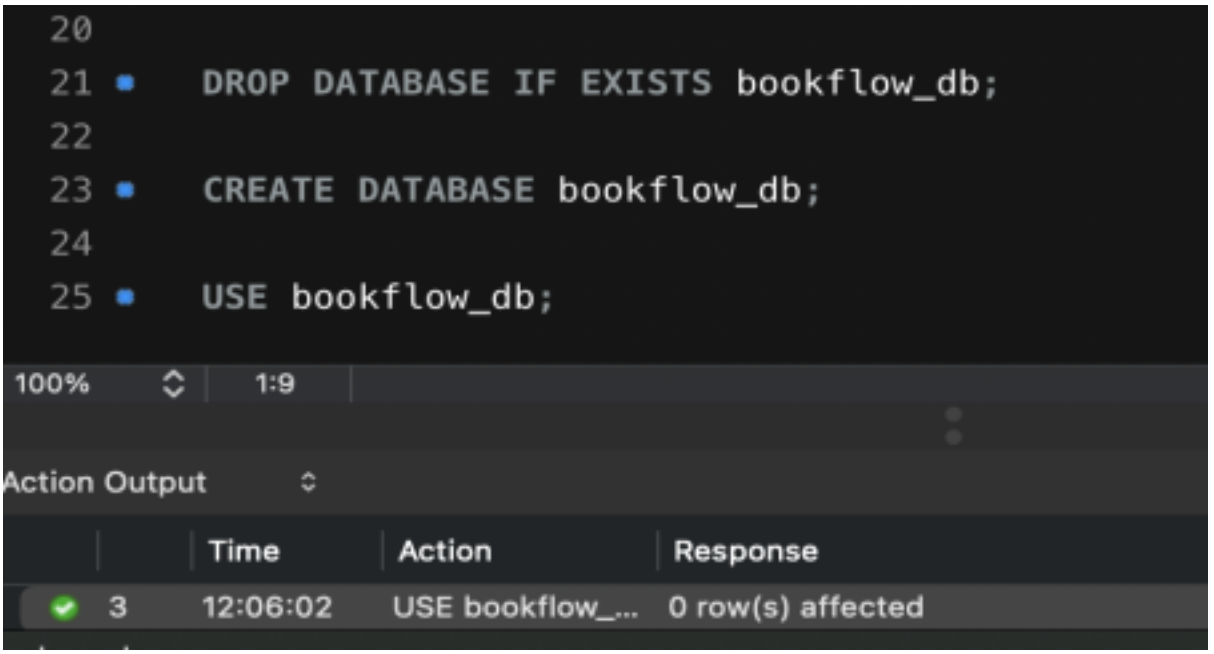
Name: vaishali

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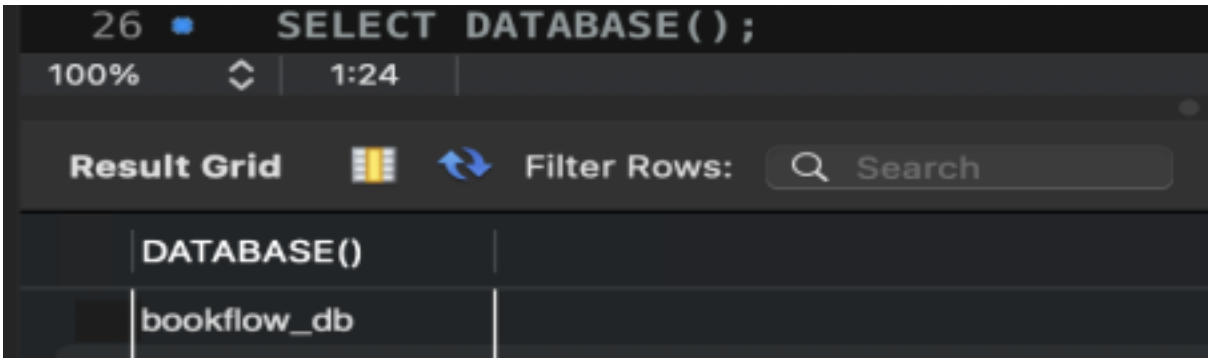
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Step 1 — Create Database

```
DROP DATABASE IF EXISTS bookflow_db;
CREATE DATABASE bookflow_db;
USE bookflow_db;
```

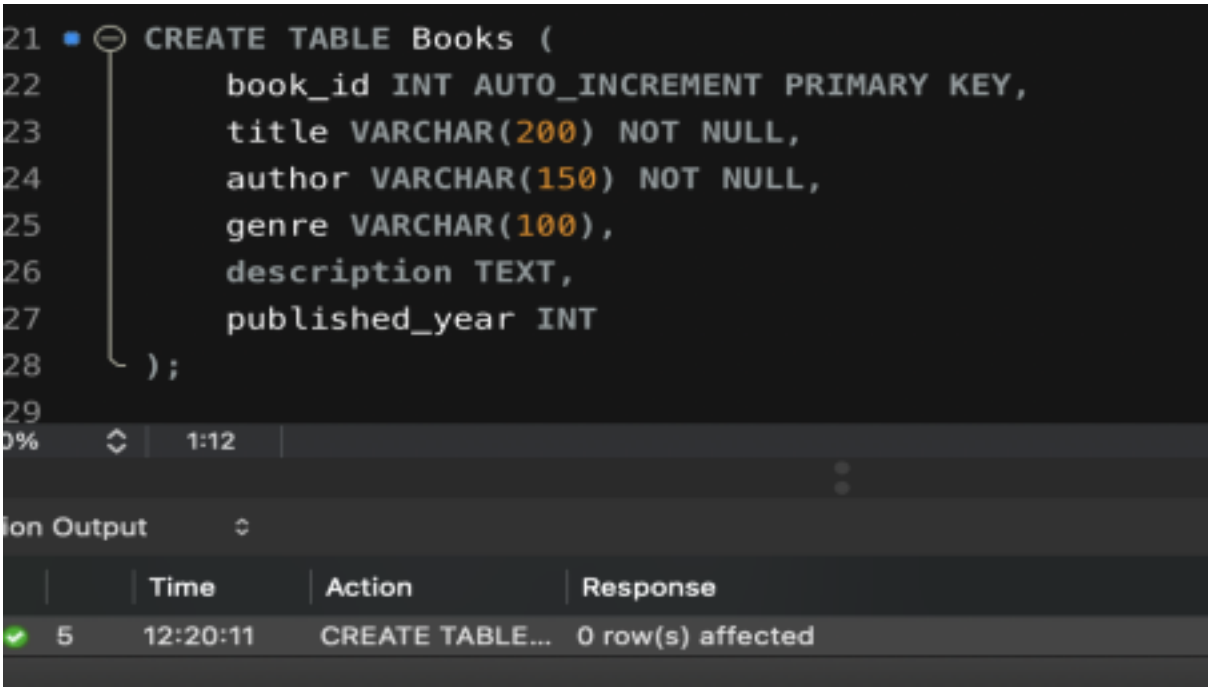



```
SELECT DATABASE();
```



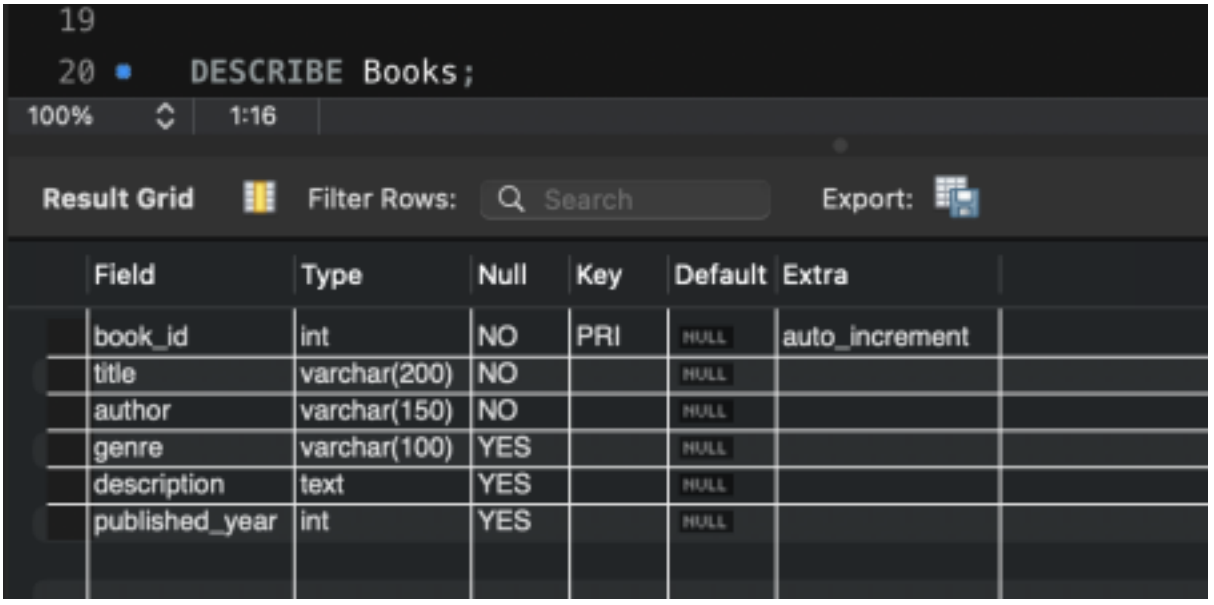
Step 2 — Create Books Table

```
CREATE TABLE Books (
    book_id INT AUTO_INCREMENT PRIMARY KEY,
    title VARCHAR(200) NOT NULL,
    author VARCHAR(150) NOT NULL,
    genre VARCHAR(100),
    description TEXT,
    published_year INT
);
```



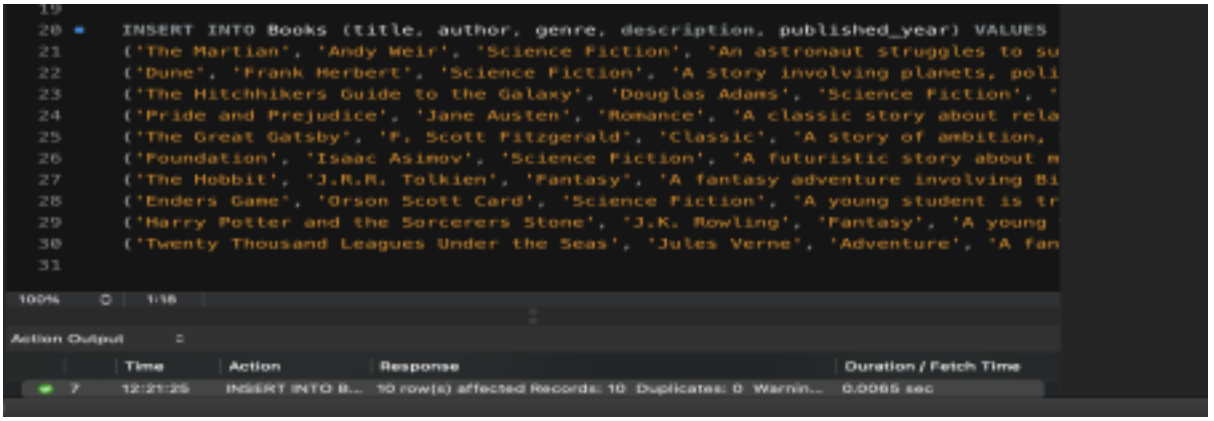
Check the structure:

```
DESCRIBE Books;
```



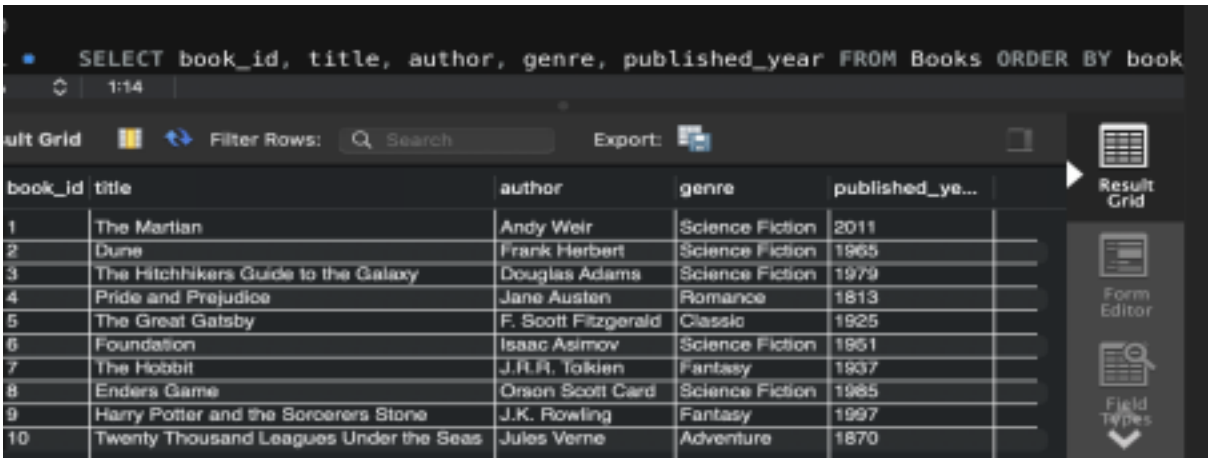
Step 3 — Insert 10 Books

```
INSERT INTO Books (title, author, genre, description, published_year) VALUES
('The Martian', 'Andy Weir', 'Science Fiction', 'An astronaut struggles to
survive alone on Mars.', 2011),
('Dune', 'Frank Herbert', 'Science Fiction', 'A story involving planets,
politics, survival and space.', 1965),
('The Hitchhikers Guide to the Galaxy', 'Douglas Adams', 'Science Fiction',
'A humorous adventure across space and alien worlds.', 1979),
('Pride and Prejudice', 'Jane Austen', 'Romance', 'A classic story
about relationships and social expectations.', 1813),
('The Great Gatsby', 'F. Scott Fitzgerald', 'Classic', 'A story of
ambition, wealth, love and the American dream.', 1925),
('Foundation', 'Isaac Asimov', 'Science Fiction', 'A futuristic story
about mathematics, civilization and space.', 1951),
('The Hobbit', 'J.R.R. Tolkien', 'Fantasy', 'A fantasy adventure involving
Bilbo Baggins and a dangerous journey.', 1937),
('Enders Game', 'Orson Scott Card', 'Science Fiction', 'A young student is
trained for an interstellar conflict.', 1985),
('Harry Potter and the Sorcerers Stone', 'J.K. Rowling', 'Fantasy', 'A young
wizard begins his magical education.', 1997),
('Twenty Thousand Leagues Under the Seas', 'Jules Verne', 'Adventure', 'A
fantastic underwater adventure aboard a mysterious submarine.', 1870);
```



Step 4 — Display All Books

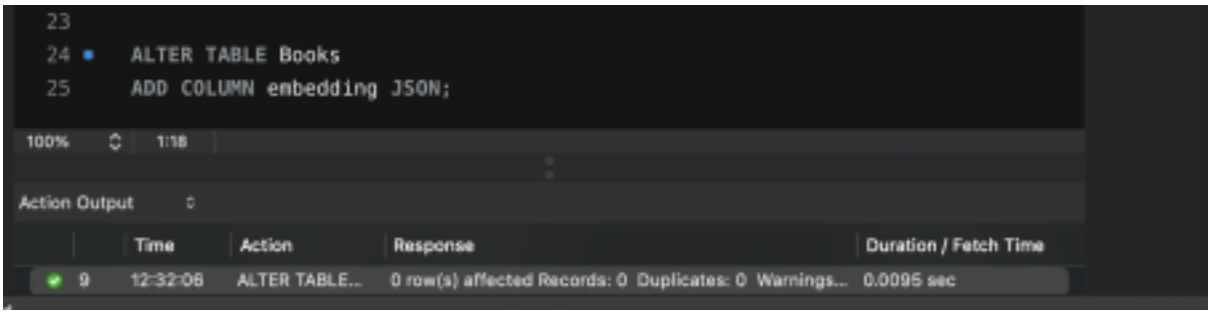
```
SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book_id;
```



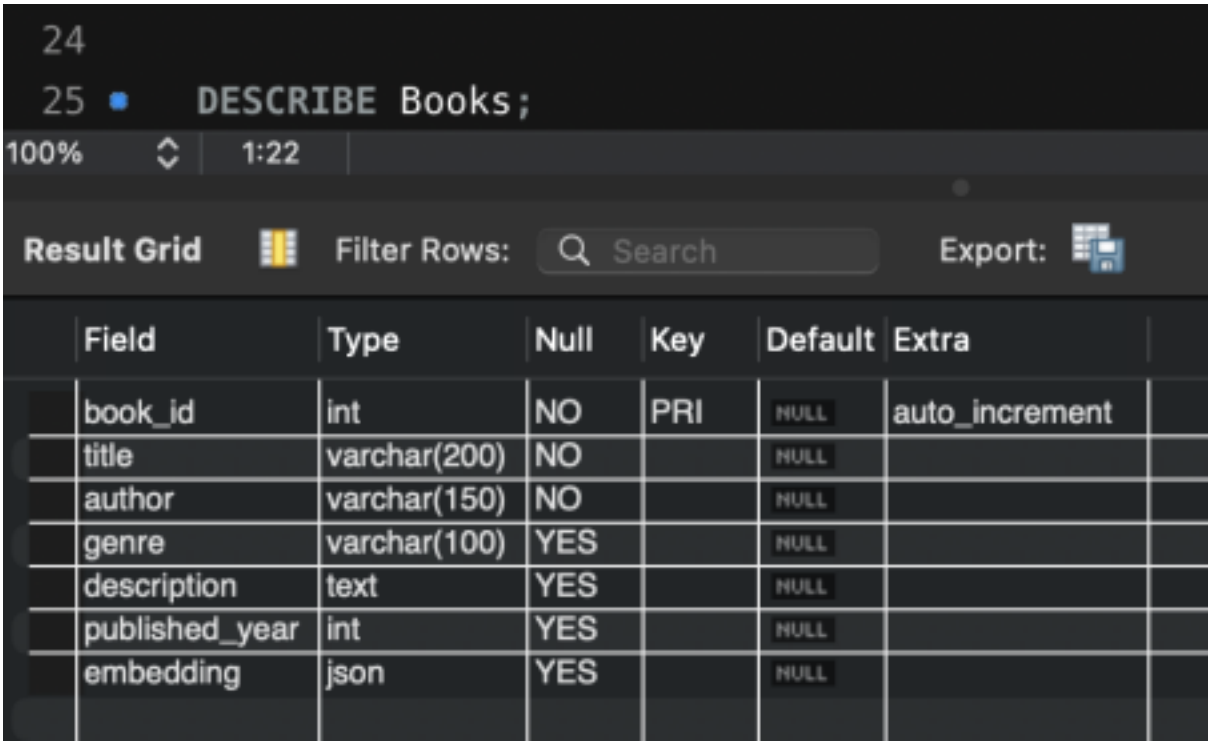
book_id	title	author	genre	published_ye...
1	The Martian	Andy Weir	Science Fiction	2011
2	Dune	Frank Herbert	Science Fiction	1965
3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	1979
4	Pride and Prejudice	Jane Austen	Romance	1813
5	The Great Gatsby	F. Scott Fitzgerald	Classic	1925
6	Foundation	Isaac Asimov	Science Fiction	1951
7	The Hobbit	J.R.R. Tolkien	Fantasy	1937
8	Enders Game	Orson Scott Card	Science Fiction	1985
9	Harry Potter and the Sorcerers Stone	J.K. Rowling	Fantasy	1997
10	Twenty Thousand Leagues Under the Seas	Jules Verne	Adventure	1870

Step 5 — Add the Embedding Column (Semantic Search Setup)

```
ALTER TABLE Books
ADD COLUMN embedding JSON;
```




```
DESCRIBE Books;
```



Step 6 — Insert Mock Vectors for Each Book

```
UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id =
1; UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id
= 2; UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE
book_id = 3; UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)
WHERE book_id = 4; UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30,
0.80) WHERE book_id = 5; UPDATE Books SET embedding = JSON_ARRAY(0.95,
0.65, 0.20) WHERE book_id = 6; UPDATE Books SET embedding =
JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7; UPDATE Books SET embedding
= JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8; UPDATE Books SET
embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9; UPDATE Books
SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10;
```

```
24
25 •  UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1;
26 •  UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2;
27 •  UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3;
28 •  UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4;
29 •  UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5;
30 •  UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6;
31 •  UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7;
32 •  UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8;
33 •  UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9;
34 •  UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10;
35
100%   1:20

Action Output
#  Time      Action                Response                                                  Duration / Fetch Time
20 12:33:52  UPDATE Books...  1 row(s) affected Rows matched: 1  Changed: 1  Warni...  0.00053 sec
```

Step 7 — Verify All Vectors

```
SELECT book_id, title, embedding FROM Books ORDER BY book_id;
```

```
22
23 •  SELECT book_id, title, embedding FROM Books ORDER BY book_id;
100%   1:13

Result Grid   Filter Rows: Search   Export:

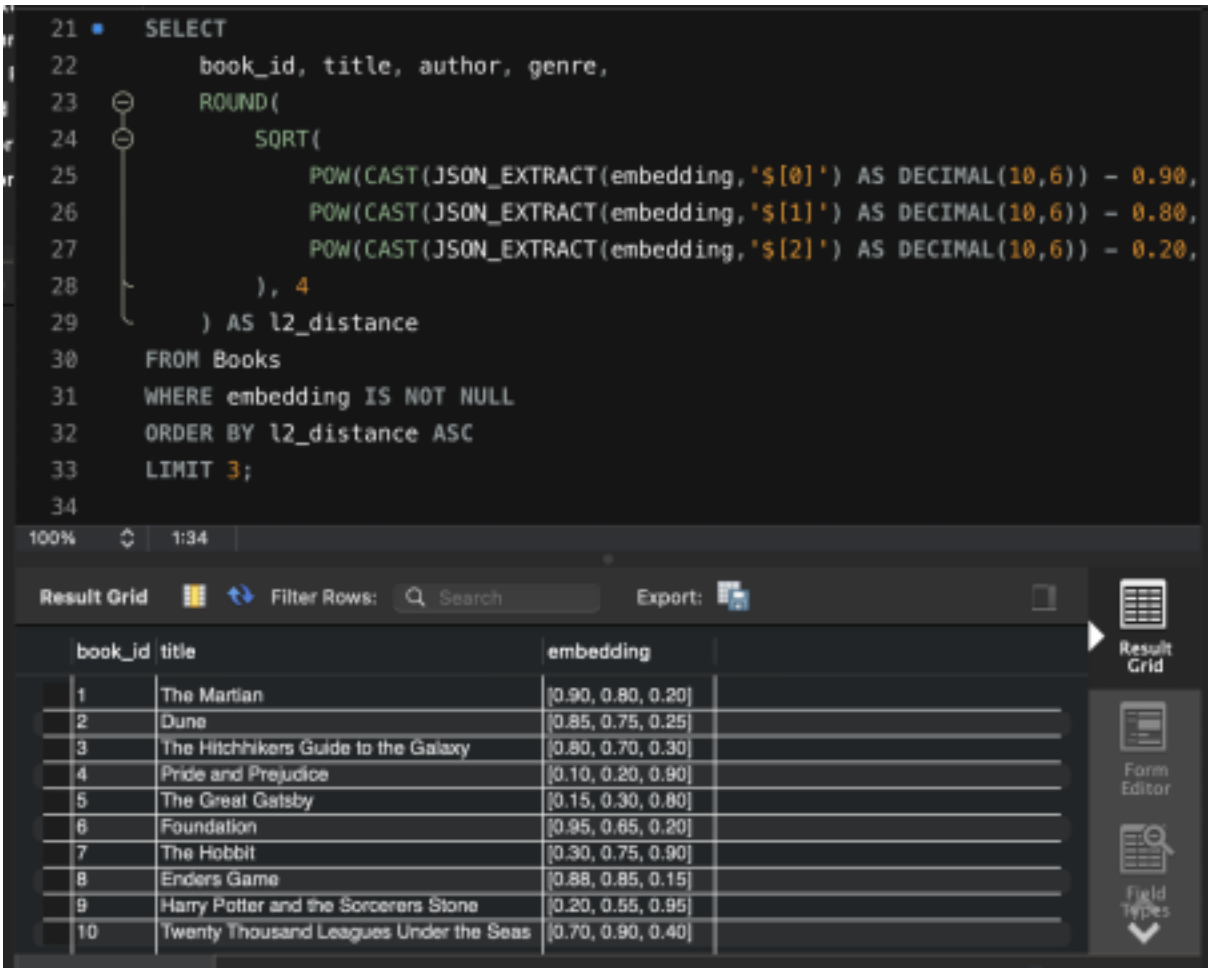
book_id | title                                | embedding
1       | The Martian                          | [0.90, 0.80, 0.20]
2       | Dune                                 | [0.85, 0.75, 0.25]
3       | The Hitchhikers Guide to the Galaxy  | [0.80, 0.70, 0.30]
4       | Pride and Prejudice                  | [0.10, 0.20, 0.90]
5       | The Great Gatsby                     | [0.15, 0.30, 0.80]
6       | Foundation                           | [0.95, 0.65, 0.20]
7       | The Hobbit                           | [0.30, 0.75, 0.90]
8       | Enders Game                          | [0.88, 0.85, 0.15]
9       | Harry Potter and the Sorcerers Stone | [0.20, 0.55, 0.95]
10      | Twenty Thousand Leagues Under the Seas | [0.70, 0.90, 0.40]
```

Step 8 — Similarity Search: L2 Distance ("closest" books)

Search vector used = [0.90, 0.80, 0.20] → represents "Stories about space". Smaller distance = more similar.

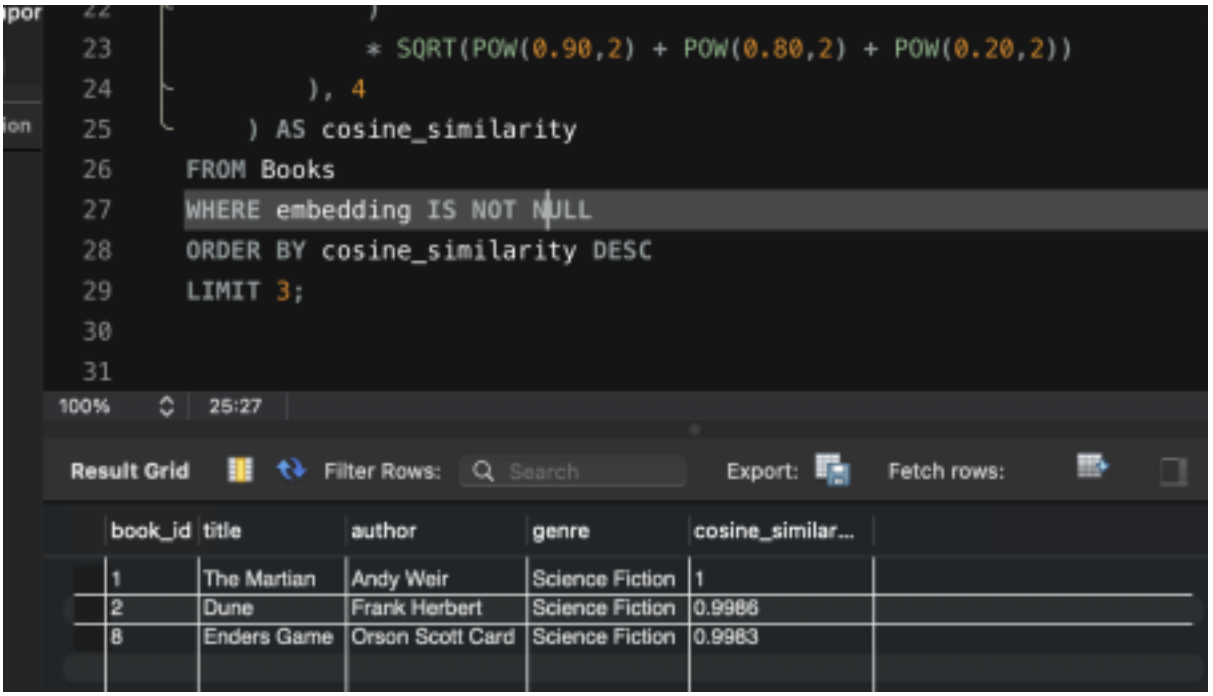
```
SELECT
    book_id, title, author, genre,
    ROUND(
        SQRT(
            POW(CAST(JSON_EXTRACT(embedding,'$[0]') AS DECIMAL(10,6)) - 0.90, 2)
        + POW(CAST(JSON_EXTRACT(embedding,'$[1]') AS DECIMAL(10,6)) - 0.80, 2)
        + POW(CAST(JSON_EXTRACT(embedding,'$[2]') AS DECIMAL(10,6)) - 0.20, 2)
        ), 4
    ) AS l2_distance
FROM Books
WHERE embedding IS NOT NULL
ORDER BY l2_distance ASC
LIMIT 3;
```

Conclusion: The 3 closest books are The Martian, Enders Game, and Dune.



Step 9 — Similarity Search: Cosine Similarity

```
SELECT
    book_id, title, author, genre,
    ROUND(
        (
            CAST(JSON_EXTRACT(embedding,'$[0]') AS DECIMAL(10,6)) * 0.90 +
            CAST(JSON_EXTRACT(embedding,'$[1]') AS DECIMAL(10,6)) * 0.80 +
            CAST(JSON_EXTRACT(embedding,'$[2]') AS DECIMAL(10,6)) * 0.20
        ) /
        (
            SQRT(
                POW(CAST(JSON_EXTRACT(embedding,'$[0]') AS DECIMAL(10,6)),2) +
                POW(CAST(JSON_EXTRACT(embedding,'$[1]') AS DECIMAL(10,6)),2) +
                POW(CAST(JSON_EXTRACT(embedding,'$[2]') AS DECIMAL(10,6)),2)
            )
            * SQRT(POW(0.90,2) + POW(0.80,2) + POW(0.20,2))
        ), 4
    ) AS cosine_similarity
FROM Books
WHERE embedding IS NOT NULL
ORDER BY cosine_similarity DESC
LIMIT 3;
```



```
            )
            * SQRT(POW(0.90,2) + POW(0.80,2) + POW(0.20,2))
        ), 4
    ) AS cosine_similarity
FROM Books
WHERE embedding IS NOT NULL
ORDER BY cosine_similarity DESC
LIMIT 3;
```

100% 25:27

Result Grid Filter Rows: Search Export: Fetch rows:

book_id	title	author	genre	cosine_similar...
1	The Martian	Andy Weir	Science Fiction	1
2	Dune	Frank Herbert	Science Fiction	0.9986
8	Enders Game	Orson Scott Card	Science Fiction	0.9983